21 A gold bar has a mass of 12.5 kg.

The density of gold is 19.3 g/cm^3

Work out the volume of the gold bar.
Give your answer correct to 3 significant figures.

12.5 kg = 12500 g. In the table, cm3 means cm^3.

	g	cm3	
	19.3	1	
$\times \frac{125000}{193}$	12500	$\frac{125000}{193}$	$\times \frac{125000}{193}$

Volume = mass ÷ density

= 12500 ÷ 19.3

= 647.668393... cm^3

2 marks total
Correct units and division

1 mark
648 cm^3 to 3 significant figures

648 cm^3

(Total for Question 21 is 3 marks)